

Research Assistant - Project Structure

1. Overview

This is a full-stack AI-powered research assistant application. It helps users generate structured research proposals through a multi-phase AI pipeline. The system is composed of three main layers: configuration (YAML), a Python/FastAPI backend, and a Next.js/TypeScript frontend.

2. Top-Level Structure

```
research-assistant/
|-- DEVELOPMENT_CALENDAR.md
|-- SETUP_GUIDE.md
|-- config/
|   |-- default.yaml
|   |-- dev.yaml
|   +-- prod.yaml
|-- backend/
|   +-- (see Section 3)
+-- frontend/
    +-- (see Section 4)
```

3. Backend Structure (Python / FastAPI)

```
backend/
|-- .env / .env.example / .gitignore
|-- main.py
|-- pyproject.toml / requirements.txt / uv.lock
|-- research_assistant.db           # SQLite DB
|-- storage/                        # File storage (empty)
|-- tests/
|   |-- integration/                # (empty)
|   +-- unit/                       # (empty)
+-- app/
    |-- main.py                     # FastAPI entrypoint
    |-- __init__.py
    |
    |-- adapters/
    |   |-- email_adapter.py
    |   |-- openai_adapter.py
    |   +-- storage_adapter.py
    |
    |-- api/
    |   |-- dependencies.py
    |   +-- routes/
    |       |-- auth.py
    |       |-- projects.py
    |       |-- research.py
    |       +-- user.py
    |
    |-- config/
    |   +-- settings.py
    |
    |-- core/
    |   |-- agents/
    |   |   |-- definitions/
    |   |   |   |-- phase0_agents.py
    |   |   |   |-- phase1_agents.py
    |   |   |   |-- phase2_agents.py
    |   |   |   |-- phase3_agents.py
    |   |   |   |-- phase4_agents.py
```

Research Assistant - Project Structure

```
| | | |-- phase5_agents.py
| | | +-- phase6_agents.py
| | +-- instructions/
| | | |-- phase0_guidelines.py
| | | |-- phase0_topic.py
| | | |-- phase1_past_projects.py
| | | |-- phase1_project_analyzer.py
| | | |-- phase1_project_finder.py
| | | |-- phase1_resource_discovery.py
| | | |-- phase2_synthesis.py
| | | |-- phase3_abstract.py
| | | |-- phase3_feasibility.py
| | | |-- phase3_justification.py
| | | |-- phase3_literature.py
| | | |-- phase3_methodology.py
| | | |-- phase3_objectives.py
| | | |-- phase3_references.py
| | | |-- phase3_timeline.py
| | | |-- phase4_formatter.py
| | | |-- phase4_orchestrator.py
| | | |-- phase5_reviewer.py
| | | +-- phase6_email.py
| |
| |-- domain/
| | | |-- project.py
| | | |-- research_spec.py
| | | |-- review.py
| | | +-- specification.py
| |
| +-- pipelines/
| | | |-- main_pipeline.py          # Orchestrator
| | | |-- phase0_workflow.py
| | | |-- phase1_workflow.py
| | | |-- phase2_workflow.py
| | | |-- phase3_workflow.py
| | | |-- phase4_workflow.py
| | | |-- phase5_workflow.py
| | | +-- phase6_workflow.py
|
|-- models/
| | |-- config.py
| | |-- database.py
| | |-- guidelines.py
| | |-- projects.py
| | |-- resources.py
| | |-- review.py
| | |-- specification.py
| | |-- synthesis.py
| | +-- user.py
|
|-- services/
| | |-- auth_service.py
| | |-- email_adapter.py
| | |-- research_service.py
| | +-- storage_service.py
|
+-- utils/
    |-- deduplication.py
    |-- file_handlers.py
    +-- save_package.py
```

4. Frontend Structure (Next.js / TypeScript)

Research Assistant - Project Structure

```
frontend/
|-- .env.local / .env.local.example
|-- next.config.js
|-- package.json / package-lock.json
|-- tailwind.config.js
|-- tsconfig.json
|-- node_modules/                # ~600+ npm packages
|-- .next/                        # Build output
|-- public/                      # Static assets (empty)
+-- src/
    |-- app/                     # Next.js App Router
    |   |-- layout.tsx
    |   |-- page.tsx             # Landing page
    |   |-- dashboard/
    |   |   |-- layout.tsx
    |   |   |-- page.tsx
    |   |   +-- projects/[id]/page.tsx
    |   |-- signin/page.tsx
    |   |-- register/page.tsx
    |   |-- forgot-password/page.tsx
    |   |-- reset-password/page.tsx
    |   +-- verify-email/page.tsx
    |
    |-- components/
    |   |-- dashboard/
    |   |   |-- Header.tsx
    |   |   |-- Sidebar.tsx
    |   |   +-- generate/page.tsx
    |   |-- generate/            # Multi-step form wizard
    |   |   |-- Step1BasicInfo.tsx
    |   |   |-- Step2FileUploads.tsx
    |   |   |-- Step3Configuration.tsx
    |   |   |-- Step4Review.tsx
    |   |   +-- StepIndicator.tsx
    |   |-- landing/
    |   |   |-- AnimatedBackground.tsx
    |   |   |-- CTA.tsx
    |   |   |-- Features.tsx
    |   |   |-- Hero.tsx
    |   |   |-- HowItWorks.tsx
    |   |   +-- Testimonials.tsx
    |   +-- ui/
    |       |-- Button.tsx
    |       +-- Card.tsx
    |
    |-- hooks/
    |   |-- useAuth.ts
    |   |-- useProjects.ts
    |   +-- useUser.ts
    |
    |-- lib/
    |   |-- api.ts
    |   +-- utils.ts
    |
    +-- types/
        +-- index.ts
```

5. Architecture Overview

Layer	Technology	Role
config/	YAML	Environment-specific configuration (default/dev/prod)

backend/	Python, FastAPI, SQLite	API server, AI pipeline, data persistence
frontend/	Next.js, TypeScript, Tailwind	UI, auth flows, dashboard